

Graphing Exponential Functions

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Identify the y-intercept, asymptote, and behavior of exponential functions of the form $y = n \cdot a^x$
- Determine whether an exponential function is increasing or decreasing based on the base and constant
- Graph exponential functions including transformations such as vertical shifts and reflections

Problems

1. Identify the y-intercept of the exponential function below:

$$y = 3^x$$

2. State whether the exponential function below is increasing or decreasing, and identify its y-intercept:

$$y = 4^{-x}$$

3. Identify the y-intercept and describe the behavior of the exponential function below:

$$y = \frac{1}{2} \cdot 5^x$$

4. Identify the y-intercept and describe the behavior of the exponential function below:

$$y = \frac{1}{3} \cdot 4^x$$

5. Identify the y-intercept, the horizontal asymptote, and describe the behavior of the function below:

$$y = 5 \cdot \left(\frac{1}{2}\right)^x$$

6. Explain why the two functions below produce the same graph:



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$$y = a^{-x} \quad \text{and} \quad y = \left(\frac{1}{a}\right)^x$$

7. Find the y-intercept and horizontal asymptote of the exponential function below, then state whether it is increasing or decreasing:

$$y = 5 \cdot 3^x - 2$$

8. Find the y-intercept and horizontal asymptote of the function below, and describe the end behavior as x increases:

$$y = -2 \cdot 4^x + 4$$

9. Compare the two exponential functions below. Identify each function's y-intercept, asymptote, and whether it is increasing or decreasing:

$$f(x) = 3 \cdot 2^x \quad \text{and} \quad g(x) = 3 \cdot \left(\frac{1}{2}\right)^x$$

10. For the exponential function below, identify the y-intercept, horizontal asymptote, and whether the graph is increasing or decreasing. Then sketch a description of the graph's shape:

$$y = -3 \cdot \left(\frac{1}{4}\right)^x + 6$$



Graphing Exponential Functions — Answer Key

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Answer Key

1. Answer: (0, 1)

- Substitute $x = 0$ into the function.
- $y = 3^0 = 1$, so the y-intercept is (0, 1).

2. Answer: Decreasing; y-intercept at (0, 1)

- When $x = 0$, $y = 4^0 = 1$, so the y-intercept is (0, 1).
- Because the exponent is negative, the function is decreasing as x increases.

3. Answer: y-intercept at (0, 1/2); increasing

- Substitute $x = 0$: $y = (1/2) \cdot 5^0 = (1/2) \cdot 1 = 1/2$.
- Since $a = 5 > 1$, the function is increasing.

4. Answer: y-intercept at (0, 1/3); increasing

- Substitute $x = 0$: $y = (1/3) \cdot 4^0 = 1/3$.
- Since $a = 4 > 1$, the function is increasing.

5. Answer: y-intercept at (0, 5); asymptote at $y = 0$; decreasing

- Substitute $x = 0$: $y = 5 \cdot 1 = 5$, so the y-intercept is (0, 5).
- Since $a = 1/2 < 1$, the function is decreasing and approaches $y = 0$ as x increases.

6. Answer: They are equivalent because $(1/a)^x = a^{-x}$

- Using exponent rules, $(1/a)^x = a^{-1 \cdot x} = a^{-x}$.
- Since the expressions are algebraically identical, their graphs are the same.

7. Answer: y-intercept at (0, 3); asymptote at $y = -2$; increasing

- Substitute $x = 0$: $y = 5 \cdot 1 - 2 = 3$, so the y-intercept is (0, 3).
- As $x \rightarrow -\infty$, $3^x \rightarrow 0$, so $y \rightarrow -2$. The horizontal asymptote is $y = -2$.
- Since $a = 3 > 1$, the function is increasing.

8. Answer: y-intercept at (0, 2); asymptote at $y = 4$; decreasing toward negative infinity

- Substitute $x = 0$: $y = -2 \cdot 1 + 4 = 2$, so the y-intercept is (0, 2).
- As $x \rightarrow -\infty$, $4^x \rightarrow 0$, so $y \rightarrow 4$. The horizontal asymptote is $y = 4$.
- As $x \rightarrow +\infty$, $-2 \cdot 4^x \rightarrow -\infty$, so the graph decreases without bound.

9. Answer: Both have y-intercept (0,3) and asymptote $y=0$; f is increasing, g is decreasing

- For both, substitute $x = 0$: $y = 3 \cdot 1 = 3$, so both have y-intercept (0, 3).
- Both approach $y = 0$ as x moves away from zero in the opposite direction.
- $f(x)$ has base $2 > 1$, so it is increasing; $g(x)$ has base $1/2 < 1$, so it is decreasing.
- The graphs are reflections of each other across the y-axis.



10. Answer: y-intercept at (0, 3); asymptote at $y = 6$; decreasing toward $y = 6$ from below

- Substitute $x = 0$: $y = -3 \cdot 1 + 6 = 3$, so the y-intercept is (0, 3).
 - As $x \rightarrow +\infty$, $(1/4)^x \rightarrow 0$, so $y \rightarrow 6$. The horizontal asymptote is $y = 6$.
 - Since the base $1/4 < 1$, the term $(1/4)^x$ decreases, making $-3 \cdot (1/4)^x$ increase toward 0.
 - Therefore the graph increases from below, approaching $y = 6$ but never reaching it.
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